

Rest area Vražné - an example of DATEX II

This example aims to demonstrate **the static** and **dynamic** part of the DATEX II publication for one particular rest.

History and purpose

This example is for the static and dynamic profile version 01/2018.

version	Author	comment
2/1/2018	Bures	Final document including description of both static and dynamic examples of truck parking information.
16/01/2018	Gelová, Bureš	Editing of the text, removal of minor typos, explanation of the concept of identifiers and editing of time records in the examples.

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1 Selected rest area

Vražné rest area stop at 319 km D1 in the direction of Prague (Brno). The rest area includes a gas station with a shop and a parking lot with equipment not only for trucks.



Picture 1 Orthophoto: Rest area Vražné with marked parking spaces

Rest area could be characterized as follows:

- Name: Vražné, location: D1 319 km direction Prague + GPS coordinates
- 1 entry and 1 exit (GPS coordinates)
- Number of places: 37x lorry (heavy vehicle), 4x bus, 6x caravan and 32x cars
- The places are spread over several rows / ensembles.
- The rest area has several services and equipment: gas station, toilet, shower, etc.

In the following sample, we do not deal with the time availability of services and equipment. The sample always presents sections of code with a comment on their content.

2 Sample static publication

The static publication is divided into logical parts (they must follow each other in the presented order - given by the scheme):

- Introductory general information
- Table information
- Rest area information
 - name and time
 - contact information
 - location
 - information about parking fees
 - services and equipment
 - parking places - groups and spaces
 - address and use of rest area
 - entry and exit from rest area
 - rest area safety
 - definition of rest area

2.1 Introductory general information

This part contains information about the provider (`supplierIdentification`) and the data creator (`publicationCreator`), which may in principle be different from the provider, as well as the type of publication (`xsi:type="GenericPublication"`) and the time it was provided (`publicationTime`).

```

1 <?xml version="1.0" encoding="UTF-8"?>
2 <d2LogicalModel modelBaseVersion="2" xmlns="http://datex2.eu/schema/2/2_0"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance">
3 <exchange>
4 <supplierIdentification>
5 <country>cz</country>
6 <nationalIdentifier>RSD CR</nationalIdentifier>
7 </supplierIdentification>
8 </exchange>
9 <payloadPublication lang="cz" xsi:type="GenericPublication">
10 <publicationTime>2017-11-20T15:25:51.153+02:00</publicationTime>
11 <publicationCreator>
12 <country>cz</country>
13 <nationalIdentifier>RSD CR</nationalIdentifier>
14 </publicationCreator>

```

2.2 Table information

This part contains the name of the publication (`genericPublicationName`) and the status (`informationStatus`) and confidentiality (`confidentiality`) of the provided data in the header. It also contains the name of the table of parking places (`parkingTableName`) "Czech Rest area Areas", the time of its creation (`parkingTableVersionTime`) and the identifier (`parkingTable@id`).

According to the international agreement, the identifier of the table starts with the two-letter abbreviation of the country (CZ) and is followed by parts specific to the country of the creator. In the case of the Czech Republic, there is the abbreviation of the purpose (ITP = intelligent truck parking) and the number of the table (01), the table must also be versioned (`parkingTable@version`).

The identifier of the table is: **CZ-ITP-01** .

```
15 <genericPublicationName>ParkingTablePublication</genericPublicationName>
16 <genericPublicationExtension>
17   <parkingTablePublication>
18     <headerInformation>
19       <confidentiality>noRestriction</confidentiality>
20       <informationStatus>real</informationStatus>
21     </headerInformation>
22     <parkingTable id="CZ-ITP-01" version="1">
23       <parkingTableName>
24         <values>
25           <value lang="en">Czech Rest area Areas</value>
26         </values>
27       </parkingTableName>
28       <parkingTableVersionTime>2016-07-
20T15:25:51.153+02:00</parkingTableVersionTime>
```

2.3 Rest area information

Each parking space (`parkingRecord`) is defined by an identifier and a version (`parkingRecord@id`, `parkingRecord@version`). Parking places can be of multiple types (`parkingRecord@xsi:type`). In this example, we are working with the non-urban parking type (`InterUrbanParkingSite`).

The identifier **CZ-01-0116** is derived from the country (CZ), the number of the rest area operator (01) and the serial number of the rest area (0116).

We do not deal with the method of assigning identifiers here; the example only shows a possible way of structuring the identifiers.

The time instant (`parkingRecordVersionTime`) refers to the date the parking record was created.

```
29 <parkingRecord id="CZ-01-0116" version="1" xsi:type="InterUrbanParkingSite">
30   <parkingName>
31     <values>
32       <value lang="cz">Vražné</value>
33     </values>
34   </parkingName>
35   <parkingRecordVersionTime>2017-11-20T15:25:51.153+02:00</parkingRecordVersionTime>
```

2.3.1 Contact information

The rest area must contain at least contact details for the operator, these details should identify local administrator so that he can be contacted in case of malfunctions. In addition to information about the operator, there can be a whole range of contact information, for example for a cleaning company, emergency services, etc.

Contact data must contain the identifier of the organization (`operator@id="CZ-01-01"`), name of the organization (`contactOrganizationName`), telephone number (`contactDetailsTelephoneNumber`) or email (`contactDetailsEMail`) and **consent to the publication of these data** (`publishingAgreement`).

```
36 <operator id="CZ-01-01" version="1" xsi:type="ContactDetails">
37   <contactOrganisationName>
38     <values>
39       <value lang="cz">Ředitelství silnic a dálnic ČR</value>
```

```

40      <value lang="en">Road and Motorway Directorate of the Czech
      Republic</value>
41    </values>
42  </contactOrganisationName>
43  <contactDetailsTelephoneNumber>+420241084111</contactDetailsTelephoneNumber>
44  <contactDetailsEMail>posta@rsd.cz</contactDetailsEMail>
45  <publishingAgreement>true</publishingAgreement>
46 </operator>

```

2.3.2 Location

The location of rest area (`parkingLocation`) is indicated by the approximate geometric center in GPS coordinates (`pointByCoordinates`). Furthermore, the position is described using the ALERT-C location (`alertCPoint`), if it exists for the given rest area (the rest area must be explicitly coded in the ALERT-C location table). Another method of describing the location is the location along the road (`pointAlongLinearElement`), containing the name, direction of the road and mileage (distance from road beginning).

The scheme also allows a description in OpenLR, but it is not listed here.

```

47 <parkingLocation xsi:type="Point">
48   <alertCPoint xsi:type="AlertCMethod2Point">
49     <alertCLocationCountryCode>2</alertCLocationCountryCode>
50     <alertCLocationTableNumber>25</alertCLocationTableNumber>
51     <alertCLocationTableVersion>6.0</alertCLocationTableVersion>
52     <alertCDirection>
53       <alertCDirectionCoded>negative</alertCDirectionCoded>
54     </alertCDirection>
55     <alertCMethod2PrimaryPointLocation>
56       <alertCLocation>
57         <specificLocation>23013</specificLocation>
58       </alertCLocation>
59     </alertCMethod2PrimaryPointLocation>
60   </alertCPoint>
61   <pointAlongLinearElement>
62     <directionRelativeAtPoint>opposite</directionRelativeAtPoint>
63     <linearElement>
64       <roadNumber>D1</roadNumber>
65     </linearElement>
66     <distanceAlongLinearElement xsi:type="DistanceFromLinearElementStart">
67       <distanceAlong>319100</distanceAlong>
68     </distanceAlongLinearElement>
69   </pointAlongLinearElement>
70   <pointByCoordinates>
71     <pointCoordinates>
72       <latitude>49.63213</latitude>
73       <longitude>17.84566</longitude>
74     </pointCoordinates>
75   </pointByCoordinates>
76 </parkingLocation>

```

2.3.3 Information about parking fees

The description of the rest area must contain information on whether parking is paid for at the given rest area or not (`tariffsAndPayment`). In the Czech Republic, there is still no payment for parking (`freeOfCharge=true`) at rest areas in the administration of the ŘSD (i.e. on the rest areas specified in the publication).

```

77 <tariffsAndPayment>
78   <freeOfCharge>true</freeOfCharge>

```

```
79 </tariffsAndPayment>
```

2.3.4 Services and facilities

It is possible to list a number of services and equipment

(`parkingEquipmentOrServiceFacility`) available at the rest area, they can be described generally for the entire rest area or specifically, for a given parking place or group of parking places.

The numbering of services and equipment (`equipmentOrServiceFacilityIndex`) starts from the value 1 and is unique within the rest area.

The most important services (`serviceFacilityType`) include: motel/hotel, restaurant, gas station, car wash and repair shop. The most important equipment (`equipmentType`) includes: car charging station, toilet, shower and internet.

Services and equipment can be further detailed by location (`groupOfLocations`) and time availability, payment requirements and accessibility for the disabled (`accessibility`).

All services and equipment must be specified as available or unavailable by (`availability`).

Note: If the service or equipment is not listed, it does not automatically mean that it is not on rest area!

Gas station (with the name of the operator)

```
80 <parkingEquipmentOrServiceFacility equipmentOrServiceFacilityIndex="1">
81   <parkingEquipmentOrServiceFacility xsi:type="ServiceFacility">
82     <availability>available</availability>
83     <nameOrBrand>
84       <values>
85         <value lang="cz">Shell</value>
86       </values>
87     </nameOrBrand>
88     <serviceFacilityType>petrolStation</serviceFacilityType>
89   </parkingEquipmentOrServiceFacility>
90 </parkingEquipmentOrServiceFacility>
```

Restaurant

```
91 <parkingEquipmentOrServiceFacility equipmentOrServiceFacilityIndex="2">
92   <parkingEquipmentOrServiceFacility xsi:type="ServiceFacility">
93     <availability>available</availability>
94     <serviceFacilityType>restaurant</serviceFacilityType>
95   </parkingEquipmentOrServiceFacility>
96 </parkingEquipmentOrServiceFacility>
```

Toilet (with coordinates and accessibility)

```
97 <parkingEquipmentOrServiceFacility equipmentOrServiceFacilityIndex="3">
98   <parkingEquipmentOrServiceFacility xsi:type="Equipment">
99     <availability>available</availability>
100     <accessibility>wheelChairAccessible</accessibility>
101     <groupOfLocations xsi:type="Point">
102       <pointByCoordinates>
103         <pointCoordinates>
104           <latitude>49.632171</latitude>
105           <longitude>17.8435696</longitude>
106         </pointCoordinates>
107       </pointByCoordinates>
108     </groupOfLocations>
109     <equipmentType>toilet</equipmentType>
```

```

110     </parkingEquipmentOrServiceFacility>
111 </parkingEquipmentOrServiceFacility>

```

2.3.5 Parking places - groups and places

The parking information must contain at least the number of parking places for trucks and refrigerated goods vehicles. Furthermore, information on special parking places (if available, e.g. for transport according to ADR).

In the example, we present the division into physical parking places (`parkingSpace`, which are identified by a serial number unique within the rest area (`parkingSpaceIndex`)). Physical parking places are further grouped into groups (`groupOfParkingSpaces`) according to location and the same characteristics. These "physical groups" are further a subset (`realSubsetOfGroup`) of statistical groups (`parkingTypeOfGroup=statisticsOnly`) indicating the total number of places for a certain type of vehicle (bus, passenger car, truck...).

For the purpose of linking groups of parking places into rows, for navigation purposes, we group physical groups using the same internal group identifier/name (`parkingSpaceOrGroupIdentifier`), e.g. (row) A.

We present groups of parking places as "not-assigned" in terms of a specific type of vehicle (`assignedParkingAmongOthers`) due to the non-enforceability / non-existence of a ban on parking e.g. cars in places for trucks. Otherwise, the group would be introduced with the type (`assignedParkingOnly`).

In the example, we first list statistical (`parkingTypeOfGroup=statisticsOnly`) and then physical groups (`parkingTypeOfGroup=adjacentSpaces`). Only physical groups contain information about the location of individual parking spaces.

Statistical group for refrigerated goods vehicles

```

112 <groupOfParkingSpaces groupIndex="1">
113   <parkingSpaceBasics xsi:type="GroupOfParkingSpaces">
114     <assignedParkingAmongOthers>
115       <vehicleCharacteristics>
116         <vehicleCharacteristicsExtension>
117           <vehicleCharacteristicsExtended>
118             <loadType2>refrigeratedGoods</loadType2>
119           </vehicleCharacteristicsExtended>
120         </vehicleCharacteristicsExtension>
121       </vehicleCharacteristics>
122     </assignedParkingAmongOthers>
123     <parkingNumberOfSpaces>0</parkingNumberOfSpaces>
124     <parkingTypeOfGroup>statisticsOnly</parkingTypeOfGroup>
125   </parkingSpaceBasics>
126 </groupOfParkingSpaces>

```

Statistical group for trucks

```

127 <groupOfParkingSpaces groupIndex="2">
128   <parkingSpaceBasics xsi:type="GroupOfParkingSpaces">
129     <assignedParkingAmongOthers>
130       <vehicleCharacteristics>
131         <vehicleType>lorry</vehicleType>
132       </vehicleCharacteristics>
133     </assignedParkingAmongOthers>
134     <parkingNumberOfSpaces>37</parkingNumberOfSpaces>
135     <parkingTypeOfGroup>statisticsOnly</parkingTypeOfGroup>

```



```

136     </parkingSpaceBasics>
137 </groupOfParkingSpaces>

```

Statistical group for buses

```

138 <groupOfParkingSpaces groupIndex="3">
139   <parkingSpaceBasics xsi:type="GroupOfParkingSpaces">
140     <assignedParkingAmongOthers>
141       <vehicleCharacteristics>
142         <vehicleType>bus</vehicleType>
143       </vehicleCharacteristics>
144     </assignedParkingAmongOthers>
145     <parkingNumberOfSpaces>4</parkingNumberOfSpaces>
146     <parkingTypeOfGroup>statisticsOnly</parkingTypeOfGroup>
147   </parkingSpaceBasics>
148 </groupOfParkingSpaces>

```

Statistical group for caravans

```

149 <groupOfParkingSpaces groupIndex="4">
150   <parkingSpaceBasics xsi:type="GroupOfParkingSpaces">
151     <assignedParkingAmongOthers>
152       <vehicleCharacteristics>
153         <vehicleType>caravan</vehicleType>
154       </vehicleCharacteristics>
155     </assignedParkingAmongOthers>
156     <parkingNumberOfSpaces>6</parkingNumberOfSpaces>
157     <parkingTypeOfGroup>statisticsOnly</parkingTypeOfGroup>
158   </parkingSpaceBasics>
159 </groupOfParkingSpaces>

```

Statistical group for passenger cars

```

160 <groupOfParkingSpaces groupIndex="5">
161   <parkingSpaceBasics xsi:type="GroupOfParkingSpaces">
162     <assignedParkingAmongOthers>
163       <vehicleCharacteristics>
164         <vehicleType>car</vehicleType>
165       </vehicleCharacteristics>
166     </assignedParkingAmongOthers>
167     <parkingNumberOfSpaces>32</parkingNumberOfSpaces>
168     <parkingTypeOfGroup>statisticsOnly</parkingTypeOfGroup>
169   </parkingSpaceBasics>
170 </groupOfParkingSpaces>

```

A physical group of 18 adjacent parking places for trucks

This group (groupOfParkingSpaces@groupIndex=10) contains information about the size of the parking places (minimumParkingSpaceDimension), the way of parking (parkingSpacePhysics and parkingMode), inclusion in the superior group (realSubsetOfGroup), the "row" assignment (parkingSpaceOrGroupIdentifier) and also the location of all 15 parking places (parkingSpace/location) and identifiers (parkingSpaceIndex); for reasons of compactness of the description, only 1 place is listed here. Parking is reserved for trucks (vehicleType=lorry).

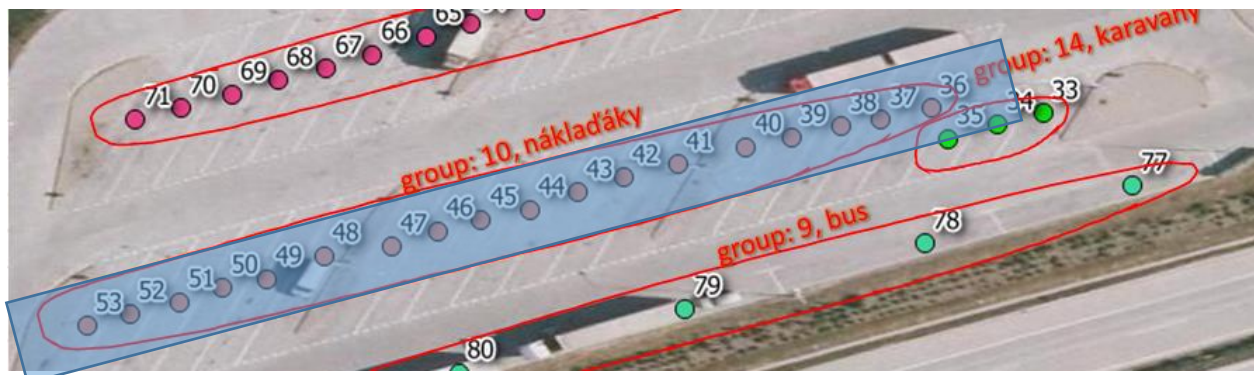


Figure 2 Orthophoto: Section of the rest area with a group of 18 parking places (36-53), reserved for trucks

```

171 <groupOfParkingSpaces groupIndex="10">
172   <parkingSpaceBasics xsi:type="GroupOfParkingSpaces">
173     <parkingSpaceOrGroupIdentifier>
174       <values>
175         <value lang="cz">Řada B</value>
176       </values>
177     </parkingSpaceOrGroupIdentifier>
178     <parkingSpacePhysics>driveThrough</parkingSpacePhysics>
179     <parkingSpacePhysics>openAir</parkingSpacePhysics>
180     <parkingMode>parallelParking</parkingMode>
181     <parkingReservation>notAvailable</parkingReservation>
182   </parkingSpaceBasics>
183   <parkingOccupancyDetectionType>singleSpaceDetection</parkingOccupancyDetectionType>
184   <assignedParkingAmongOthers>
185     <vehicleCharacteristics>
186       <vehicleType>lorry</vehicleType>
187     </vehicleCharacteristics>
188   </assignedParkingAmongOthers>
189   <parkingNumberOfSpaces>18</parkingNumberOfSpaces>
190   <parkingTypeOfGroup>adjacentSpaces</parkingTypeOfGroup>
191   <realSubsetOfGroup>2</realSubsetOfGroup>
192   <minimumParkingSpaceDimension>
193     <dimensionLength>25</dimensionLength>
194     <dimensionWidth>5</dimensionWidth>
195   </minimumParkingSpaceDimension>
196   <parkingSpace parkingSpaceIndex="36">
197     <location xsi:type="Point">
198       <pointByCoordinates>
199         <pointCoordinates>
200           <latitude>49.631545</latitude>
201           <longitude>17.8420908</longitude>
202         </pointCoordinates>
203       </pointByCoordinates>
204     </location>
205   </parkingSpace>
206 </parkingSpace>
207 </parkingSpaceBasics>
208 </groupOfParkingSpaces>

```

A physical group of three adjacent caravan parking places

This group contains information about caravan parking places (`vehicleType=caravan`) in a similar structure to the previous group, and since it is in the same row as the previous one, it has the same group name (`parkingSpaceOrGroupIdentifier`). Parking places have a serial

identifier (parkingSpaceIndex) unique within the scope of one rest area. Again, only 1 parking place is shown here for reasons of compactness of the description.

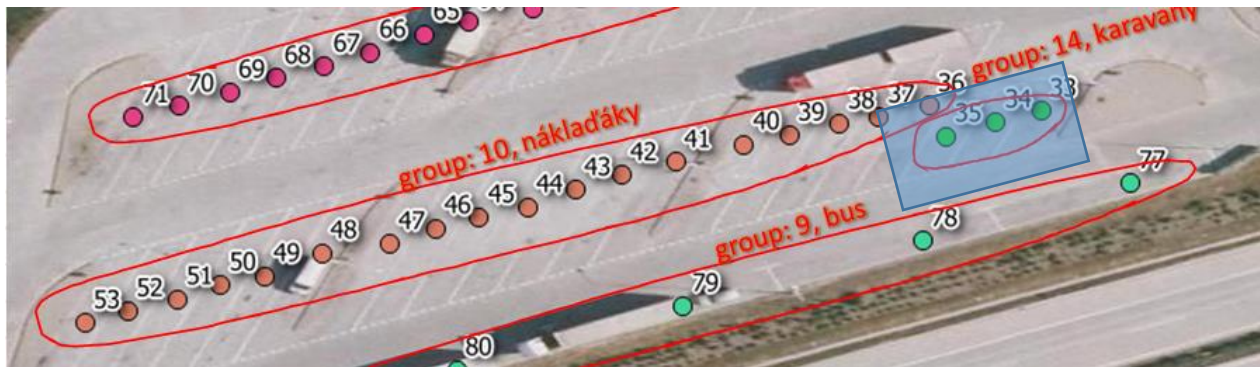


Figure 3 Orthophoto: Section of the rest area with a group of 3 parking places (33-35), reserved for caravans

```

209 <groupOfParkingSpaces groupId="14">
210   <parkingSpaceBasics xsi:type="GroupOfParkingSpaces">
211     <parkingSpaceOrGroupIdentifier>
212       <values>
213         <value lang="cz">Řada B</value>
214       </values>
215     </parkingSpaceOrGroupIdentifier>
216     <parkingSpacePhysics>driveThrough</parkingSpacePhysics>
217     <parkingSpacePhysics>openAir</parkingSpacePhysics>
218     <parkingMode>parallelParking</parkingMode>
219     <parkingReservation>notAvailable</parkingReservation>
220   </parkingSpaceBasics>
221   <parkingOccupancyDetectionType>singleSpaceDetection</parkingOccupancyDetectionType>
222   <assignedParkingAmongOthers>
223     <vehicleCharacteristics>
224       <vehicleType>caravan</vehicleType>
225     </vehicleCharacteristics>
226   </assignedParkingAmongOthers>
227   <parkingNumberOfSpaces>3</parkingNumberOfSpaces>
228   <parkingTypeOfGroup>adjacentSpaces</parkingTypeOfGroup>
229   <realSubsetOfGroup>2</realSubsetOfGroup>
230   <minimumParkingSpaceDimension>
231     <dimensionLength>15</dimensionLength>
232     <dimensionWidth>5</dimensionWidth>
233   </minimumParkingSpaceDimension>
234   <parkingSpace parkingSpaceIndex="33">
235     <parkingSpace>
236       <location xsi:type="Point">
237         <pointByCoordinates>
238           <pointCoordinates>
239             <latitude>49.6315382</latitude>
240             <longitude>17.8422427</longitude>
241           </pointCoordinates>
242         </pointByCoordinates>
243       </location>
244     </parkingSpace>
245   </parkingSpace>
246 </groupOfParkingSpaces>

```

2.3.6 Address and use of rest area

Rest area address (`parkingSiteAddress`) is also required; however, if it is located on the highway, it usually does not have an address, so it is listed here as undefined (`contactNotDefined=true`).

```
247 <parkingSiteAddress xsi:type="Contact">
248   <contactNotDefined>true</contactNotDefined>
249 </parkingSiteAddress>
```

Parking usage scenario (`parkingUsageScenario`) is also required; the latter refers to the primary purpose of the rest area, i.e. truck parking (`parkingUsageScenario=truckParking`).

```
250 <parkingUsageScenario scenarioIndex="1">
251   <parkingUsageScenario>
252     <parkingUsageScenario>truckParking</parkingUsageScenario>
253     <truckParkingDynamicManagement>noDynamicParkingManagement</truckParkingDynamicManag
254     </parkingUsageScenario>
255   </parkingUsageScenario>
```

2.3.7 Entry and exit from the rest area

It is necessary to indicate at least one real entrance and exit from the rest area (`parkingAccess`), it is not the highway exit. The entry is given by the coordinate (`Location/pointByCoordinates`), the main related road (`primaryRoad`), its direction (`roadDestination`) and the distance of the entry from the road (`distanceToThisRoad`). The entry/exit identifier (`parkingAccess@id="CZ-01-0116-A001`) is chosen to be globally unique, i.e. the rest area number and the entry/exit sequence number. This allows information to be reused between rest areas.

The rest area is located directly on the highway (`distanceToThisRoad=0`).

Rest area entry

```
256 <parkingAccess id="CZ-01-0116-A001">
257   <accessCategory>vehicleEntrance</accessCategory>
258   <primaryRoad>
259     <roadIdentifier>
260       <values>
261         <value lang="cz">D1</value>
262       </values>
263     </roadIdentifier>
264     <typeOfRoad>motorway</typeOfRoad>
265     <roadDestination>
266       <values>
267         <value lang="cz">Bělotín</value>
268       </values>
269     </roadDestination>
270     <distanceToThisRoad>0</distanceToThisRoad>
271   </primaryRoad>
272   <location xsi:type="Point">
273     <pointByCoordinates>
274       <pointCoordinates>
275         <latitude>49.63213</latitude>
276         <longitude>17.84566</longitude>
277       </pointCoordinates>
278     </pointByCoordinates>
279   </location>
```

```
280 </parkingAccess>
```

Rest area exit

```
281 <parkingAccess id="CZ-01-0116-A002">
282   <accessCategory>vehicleExit</accessCategory>
283   <primaryRoad>
284     <roadIdentifier>
285       <values>
286         <value lang="cz">D1</value>
287       </values>
288     </roadIdentifier>
289     <typeOfRoad>motorway</typeOfRoad>
290     <roadDestination>
291       <values>
292         <value lang="cz">Bělotín</value>
293       </values>
294     </roadDestination>
295     <distanceToThisRoad>0</distanceToThisRoad>
296   </primaryRoad>
297   <location xsi:type="Point">
298     <pointByCoordinates>
299       <pointCoordinates>
300         <latitude>49.630960</latitude>
301         <longitude>17.840672</longitude>
302       </pointCoordinates>
303     </pointByCoordinates>
304   </location>
305 </parkingAccess>
```

2.3.8 Safety and rest area usage

Rest areas for truck parking must contain information about passive (fences, lights) and active (cameras, guard, dog...) security (parkingStandardsAndSecurity).

A rest area on an interurban parking lot (interUrbanParkingSiteLocation) must have the type of road to which it belongs defined.

```
306 <parkingStandardsAndSecurity>
307   <labelSecurityLevel>none</labelSecurityLevel>
308   <parkingSecurity>lighting</parkingSecurity>
309   <parkingSecurity>fences</parkingSecurity>
310 </parkingStandardsAndSecurity>
311 <interUrbanParkingSiteLocation>motorway</interUrbanParkingSiteLocation>
```

2.4 Conclusion of publication

A publication can contain N information about rests areas in M tables. This is followed by the end of the publication.

```
312 </parkingRecord>
313   </parkingTable>
314 </parkingTablePublication>
315 </genericPublicationExtension>
316 </payloadPublication>
317 </d2LogicalModel>
```

3 Example of a dynamic publication

The dynamic publication is used to inform about changes in the static publication and the occupancy status of the rest area / parking places. Semi-static changes that the dynamic publication transmits are changes to services and facility, their tariffs, availability, etc.

All static parts to which dynamic content is published are referenced to the static publication by a specific identifier of the table, rest area and parking zone number, place or facility or service.

A dynamic publication consists of the following parts:

- Introductory general information
- Information about the referenced table
- Rest area status information
 - time of publication
 - parking place and zone occupancy statuses
 - changes in the status of equipment and services
 - general status of rest area

3.1 Introductory general information

This part contains information about the provider (`supplierIdentification`) and the data creator (`publicationCreator`), which in principle can be different from the provider, as well as the type of publication (`xsi:type="GenericPublication"`) and the time it was provided (`publicationTime`).

```
1 <?xml version="1.0" encoding="UTF-8"?>
2 <d2LogicalModel modelBaseVersion="2" xmlns="http://datex2.eu/schema/2/2_0"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance">
3   <exchange>
4     <supplierIdentification>
5       <country>cz</country>
6       <nationalIdentifier>RSD CR</nationalIdentifier>
7     </supplierIdentification>
8   </exchange>
9   <payloadPublication lang="cz" xsi:type="GenericPublication">
10    <publicationTime>2017-12-22T09:05:29.0</publicationTime>
11    <publicationCreator>
12      <country>cz</country>
13      <nationalIdentifier>RSD CR</nationalIdentifier>
14    </publicationCreator>
```

3.2 Reference to the table of parking spaces

This section contains the name of the publication (`ParkingStatusPublication`) and the status (`informationStatus`) and confidentiality (`confidentiality`) of the provided data in the header. It also contains a reference to the static publication, or to the used table of parking places to which the statuses in this publication refer (`parkingTableReference`), its identifier (`id`) and version (`version`).

```
15 <genericPublicationName>ParkingStatusPublication</genericPublicationName>
16 <genericPublicationExtension>
17   <parkingStatusPublication>
18     <headerInformation>
```

```

19      <confidentiality>noRestriction</confidentiality>
20      <informationStatus>real</informationStatus>
21    </headerInformation>
22    <parkingTableReference id="CZ-ITP-01" targetClass="ParkingTable"
version="1"/>

```

3.3 Rest area status information

Each parking status information (`parkingRecordStatus`) is defined in the dynamic publication by a reference to the source static parking information (`parkingRecordReference`) by its identifier and version.

Identifier **CZ-01-0116** refers to the rest area in a static publication.

The timestamp (`parkingStatusOriginTime`) refers to the date the parking status message was created.

```

23 <parkingRecordStatus xsi:type="ParkingSiteStatus">
24   <parkingRecordReference id="CZ-01-0116" targetClass="ParkingRecord" version="1"/>
25   <parkingStatusOriginTime>2017-12-22T09:05:29.0</parkingStatusOriginTime>

```

3.3.1 Parking places occupancy statuses

Information about the status of a parking place (`parkingSpaceStatus`) is referenced to the static publication using an index (`parkingSpaceIndex`) and presented in a flat structure, not "encapsulated" in the corresponding group of parking spaces. The statuses of parking places (`parkingSpaceOccupied`) are listed as occupied or free for all parking places for which the status is known.

Note: If the status of the parking place is not provided it means that the status is unknown.

Here, for simplicity, we only list the status of 2 parking places and not all; this part occupies 90% of the real publication.

Parking places statuses

```

26 <parkingSpaceStatus parkingSpaceIndex="1">
27   <parkingSpaceStatus>
28     <parkingSpaceOccupied>false</parkingSpaceOccupied>
29   </parkingSpaceStatus>
30 </parkingSpaceStatus>
31 <parkingSpaceStatus parkingSpaceIndex="2">
32   <parkingSpaceStatus>
33     <parkingSpaceOccupied>true</parkingSpaceOccupied>
34   </parkingSpaceStatus>
35 </parkingSpaceStatus>

```

3.3.2 Status of the rest area

The status of the rest area as a whole (`parkingOccupancy`) consists of the number of free places (`parkingNumberOfVacantSpaces`), the number of occupied places (`parkingNumberOfOccupiedSpaces`), the number of parked vehicles (`parkingNumberOfVehicles`) and the trend (`parkingOccupancyTrend`), **regardless of type**.

All parts are needed; their combination gives information about the number of parked vehicles outside the monitored places and the number of wrongly parked vehicles (across more parking places).


```

36 <parkingOccupancy>
37   <parkingNumberOfVacantSpaces>60</parkingNumberOfVacantSpaces>
38   <parkingNumberOfOccupiedSpaces>20</parkingNumberOfOccupiedSpaces>
39   <parkingNumberOfVehicles>16</parkingNumberOfVehicles>
40   <parkingOccupancyTrend>stable</parkingOccupancyTrend>
41 </parkingOccupancy>

```

3.3.3 Statuses of groups of parking places

Statuses of groups of parking places is an additional to information to the statuses of physical parking places. It can be derived from.

Parking statuses are also present for individual physical and statistical groups (groupOfParkingSpacesStatus); the reason is the same as the reason for rest area status. Disproportions indicate parking problems (poorly parked vehicles, vehicles parked beyond the capacity of the group, etc.)

Group 1 for frozen goods vehicles is not listed here, because in the static publication this group has 0 parking places at this rest area, so no vehicles can be expected in this group. We list the other groups of parking spaces.

Note: If the status of the group of parking places is not provided it means an unknown occupancy status.

Information for statistical group 2 (trucks)

Occupied parking places (parkingNumberOfOccupiedSpaces) do not correspond to the number of parked vehicles (parkingNumberOfVehicles).

```

42 <groupOfParkingSpacesStatus groupIndex="2">
43   <groupOfParkingSpacesStatus>
44     <parkingNumberOfVacantSpaces>28</parkingNumberOfVacantSpaces>
45     <parkingNumberOfOccupiedSpaces>9</parkingNumberOfOccupiedSpaces>
46     <parkingNumberOfVehicles>5</parkingNumberOfVehicles>
47   </groupOfParkingSpacesStatus>
48 </groupOfParkingSpacesStatus>

```

Information for statistical group 3 (buses)

The parking places in the bus group are filled with trucks, but this is irrelevant from the point of view of reporting.

```

49 <groupOfParkingSpacesStatus groupIndex="3">
50   <groupOfParkingSpacesStatus>
51     <parkingNumberOfVacantSpaces>1</parkingNumberOfVacantSpaces>
52     <parkingNumberOfOccupiedSpaces>3</parkingNumberOfOccupiedSpaces>
53   </groupOfParkingSpacesStatus>
54 </groupOfParkingSpacesStatus>

```

Information for real groups 10 (trucks) and 14 (caravans) in a row B

1 van and 1 truck (longitudinally) are parked in this group, occupying 5 transverse spaces (see spaces 36-40 in the image below). In the case of disproportions between the number of detected occupied spaces and the actual number of parked vehicles, the parameter (parkingNumberOfVehicles) is provided, but it can sometimes be difficult to detect.

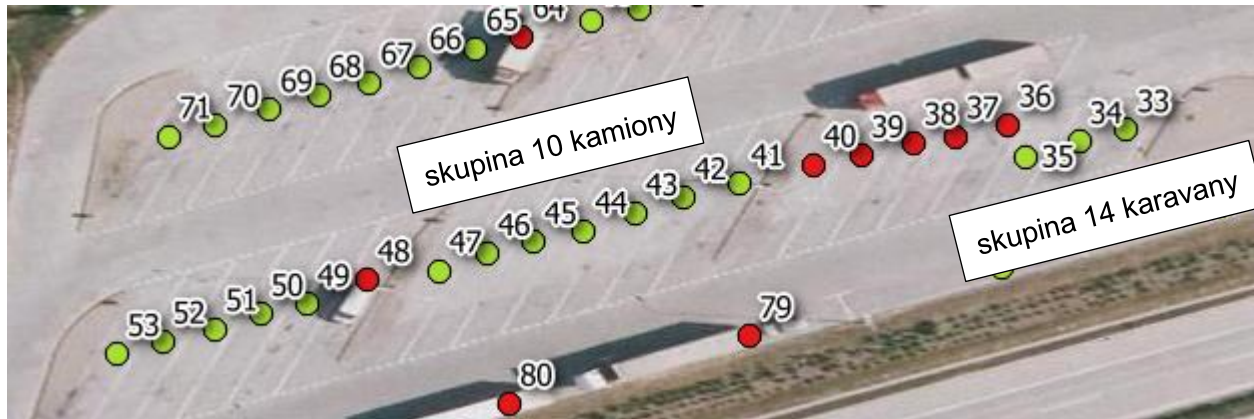


Figure 4 Orthophoto: Section of the rest area with a group of parking places 10 and 14, reserved for trucks and caravans (red means occupied)

```

55 <groupOfParkingSpacesStatus groupIndex="10">
56   <groupOfParkingSpacesStatus>
57     <parkingNumberOfVacantSpaces>12</parkingNumberOfVacantSpaces>
58     <parkingNumberOfOccupiedSpaces>6</parkingNumberOfOccupiedSpaces>
59     <parkingNumberOfVehicles>2</parkingNumberOfVehicles>
60   </groupOfParkingSpacesStatus>
61 </groupOfParkingSpacesStatus>
62 <groupOfParkingSpacesStatus groupIndex="14">
63   <groupOfParkingSpacesStatus>
64     <parkingNumberOfVacantSpaces>3</parkingNumberOfVacantSpaces>
65     <parkingNumberOfOccupiedSpaces>0</parkingNumberOfOccupiedSpaces>
66   </groupOfParkingSpacesStatus>
67 </groupOfParkingSpacesStatus>

```

3.3.4 Changes to the status of facilities and services

For each indexed facility or service, a change in the number of facilities and sub-items (number of racks at a gas station), the number of free facilities or facility items, opening hours or operational status can be reported (using `parkingEquipmentOrServiceFacilityStatus`). Below we report an inaccessible (broken) toilet (`equipmentOperationStatus`), listed on this rest area as equipment with index 3.

```

68 <parkingEquipmentOrServiceFacilityStatus equipmentOrServiceFacilityIndex="3">
69   <parkingEquipmentOrServiceFacilityStatus>
70     <equipmentOperationStatus>notInOperation</equipmentOperationStatus>
71   </parkingEquipmentOrServiceFacilityStatus>
72 </parkingEquipmentOrServiceFacilityStatus>

```

3.3.5 General status of rest area

For each rest area, we list its status below. Whether it is open (`parkingSiteOpeningStatus`) and whether parking places are available (`parkingSiteStatus`).

```

73 <parkingSiteStatus>spacesAvailable</parkingSiteStatus>
74 <parkingSiteOpeningStatus>open</parkingSiteOpeningStatus>

```

3.4 Conclusion of publication

Information about other N rests from M rest area tables can also follow. This is followed by the end of the publication.

```
75         </parkingRecordStatus>
76     </parkingStatusPublication>
77 </genericPublicationExtension>
78 </payloadPublication>
79 </d2LogicalModel>
```